Teoría general de osciladores aplicada al caso: 

$$L = \frac{1}{2} m \dot{x}_1^2 + \frac{1}{2} m \dot{x}_2^2 - \frac{1}{2} k x_1^2 - \frac{1}{2} k x_2^2 - \frac{1}{2} k_{12} (x_2 - x_1)^2$$

→ Coordenadas generalizadas: $q_1 = x_1$; $q_2 = x_2 \Rightarrow \vec{q} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

→ Ecuación de movimiento: $\{m\} \ddot{\vec{q}} + \{A\} \vec{q} = 0$

⇒ Matriz $\{m\}$:

$$T = \frac{1}{2} \sum_{j,k=1}^{2} m_{jk} \dot{q}_j \dot{q}_k = \frac{1}{2} m_{11} \dot{q}_1^2 + \frac{1}{2} m_{12} \dot{q}_1 \dot{q}_2 + \frac{1}{2} m_{21} \dot{q}_2 \dot{q}_1 + \frac{1}{2} m_{22} \dot{q}_2^2 =$$

$$= \frac{1}{2} m_{11} \dot{x}_1^2 + \frac{1}{2} m_{12} \dot{x}_1 \dot{x}_2 + \frac{1}{2} m_{21} \dot{x}_1 \dot{x}_2^2 + \frac{1}{2} m_{22} \dot{x}_2^2 =$$

De L →

$$= \frac{1}{2} m \dot{x}_1 + \frac{1}{2} m \dot{x}_2^2 \Rightarrow$$

$$\Rightarrow \begin{cases} m_{11} = m \\ m_{12} = m_{21} = 0 \\ m_{22} = m \end{cases} \Rightarrow \{m\} = \begin{pmatrix} m & 0 \\ 0 & m \end{pmatrix}$$

⇒ Matriz $\{A\}$:

$$V = \frac{1}{2} \sum_{j,k=1}^{2} A_{jk} q_j q_k \; ; \quad A_{jk} = \frac{\partial^2 V}{\partial q_j \partial q_k} \; ; \quad V = \frac{1}{2} k x_1^2 + \frac{1}{2} k x_2^2 + \frac{1}{2} k_{12} (x_2 - x_1)^2$$

$$\begin{cases} A_{11} = \dfrac{\partial^2 V}{\partial x_1^2} = k + k_{12} \\ A_{22} = \dfrac{\partial^2 V}{\partial x_2^2} = k + k_{12} \\ A_{12} = A_{21} = \dfrac{\partial^2 V}{\partial x_1 \partial x_2} = -k_{12} \end{cases} \Rightarrow \{A\} = \begin{pmatrix} k + k_{12} & -k_{12} \\ -k_{12} & k + k_{12} \end{pmatrix}$$

$$\begin{cases} \dfrac{\partial V}{\partial x_1} = k x_1 - k_{12} (x_2 - x_1) \\ \dfrac{\partial V}{\partial x_2} = k x_2 + k_{12} (x_2 - x_1) \end{cases}$$

→ Frecuencias características:

$$\det\left[\{A\} - \omega^2\{m\}\right] = 0 \Rightarrow \begin{vmatrix} k+k_{12}-\omega^2 m & -k_{12} \\ -k_{12} & k+k_{12}-\omega^2 m \end{vmatrix} = 0 \Rightarrow$$

Ecuación secular:

$$\Rightarrow (k+k_{12}-\omega^2 m)^2 = k_{12}^2 \Rightarrow k+k_{12}-\omega^2 m = \pm k_{12} \Rightarrow$$

$$\Rightarrow \begin{cases} k+k_{12}-\omega^2 m = k_{12} \Rightarrow \omega^2 = \frac{k}{m} \Rightarrow \omega = \omega_S = \sqrt{\frac{k}{m}} \\ k+k_{12}-\omega^2 m = -k_{12} \Rightarrow \omega^2 = \frac{k+2k_{12}}{m} \Rightarrow \omega = \omega_A = \sqrt{\frac{k+2k_{12}}{m}} \end{cases}$$

→ Modos y coordenadas normales:

$$\left.\begin{aligned} \eta_1(t) &= A_S \cos(\omega_S t - \delta_S) \\ \eta_2(t) &= A_A \cos(\omega_A t - \delta_A) \end{aligned}\right\}$$ cualquier solución es combinación lineal de estas soluciones. ⇒

$$\Rightarrow \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}$$ donde $\vec{a}_1 = \begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix}$ y $\vec{a}_2 = \begin{pmatrix} a_{21} \\ a_{22} \end{pmatrix}$

verifican que $\left[\{A\} - \omega_n^2 \{m\}\right] \vec{a}_n = 0$

Modos normales Obtengamos ahora los modos normales del sistema.

Modo $n=1$ $\omega_1 = \sqrt{k/m}$ y el sistema oscila de tal manera que $\eta_2(t) = 0 \ \forall t$

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{21} \end{pmatrix} \begin{pmatrix} \eta_1 \\ 0 \end{pmatrix} \Rightarrow \left.\begin{aligned} x_1 &= a_{11}\eta_1 \\ x_2 &= a_{21}\eta_1 \end{aligned}\right\} \Rightarrow$$ sólo nos hacen falta a_{11} y a_{21} es decir $\vec{a}_1$

para $\vec{a}_1$ verifica: $\omega_1 = \sqrt{\frac{k}{m}}$

$$\left[\{A\} - \omega_1^2 \{m\}\right] \vec{a}_1 = 0 \Rightarrow \begin{pmatrix} k+k_{12}-\frac{k}{m}m & -k_{12} \\ -k_{12} & k+k_{12}-\frac{k}{m}m \end{pmatrix} \begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix} = 0$$

$$\Rightarrow \begin{pmatrix} k_{12} & -k_{12} \\ -k_{12} & k_{12} \end{pmatrix}\begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} k_{12}a_{11} - k_{12}a_{12} = 0 \\ -k_{12}a_{11} + k_{12}a_{21} = 0 \end{cases} \Rightarrow \boxed{a_{11} = a_{21}} \Rightarrow$$

$$\Rightarrow \vec{a}_1 = \begin{pmatrix} a \\ a \end{pmatrix} \Rightarrow \begin{cases} x_1 = a\eta_1 \\ x_2 = a\eta_1 \end{cases} \Rightarrow \text{misma amplitud ("a") y con el mismo signo.}$$

$$\hookrightarrow \boxed{\frac{x_1}{x_2} = 1}$$

$n = 2$ $\omega_2 = \sqrt{\frac{k+k_{12}}{m}}$ y el sistema oscila de tal manera que $\eta_1 = 0 \ \forall t$

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}\begin{pmatrix} 0 \\ \eta_2 \end{pmatrix} \Rightarrow \begin{cases} x_1 = a_{12}\eta_2 \\ x_2 = a_{22}\eta_1 \end{cases} \Rightarrow \text{vector } \vec{a}_2$$

pero $\vec{a}_2$ verifica: $\omega_2 = \sqrt{\frac{k+2k_{12}}{m}}$

$$\left[\,|A| - \omega_2^2\,|m|\,\right]\vec{a}_2 = 0 \Rightarrow \begin{pmatrix} k + k_{12} - \frac{k+2k_{12}}{m}m & -k_{12} \\ -k_{12} & k + k_{12} - \frac{k+k_{12}}{m}m \end{pmatrix}\begin{pmatrix} a_{21} \\ a_{22} \end{pmatrix} = 0$$

$$\Rightarrow \begin{pmatrix} -k_{12} & -k_{12} \\ -k_{12} & -k_{12} \end{pmatrix}\begin{pmatrix} a_{12} \\ a_{22} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} -k_{12}a_{12} - k_{12}a_{22} = 0 \\ -k_{12}a_{12} - k_{12}a_{22} = 0 \end{cases} \Rightarrow$$

$$\Rightarrow a_{12} = -a_{22} \rightarrow \vec{a}_2 = \begin{pmatrix} \tilde{a} \\ -\tilde{a} \end{pmatrix} \Rightarrow \begin{cases} x_1 = \tilde{a}\eta_2 \\ x_2 = -\tilde{a}\eta_2 \end{cases} \Rightarrow \text{misma amplitud y con signos contrario}$$

$$\hookrightarrow \boxed{\frac{x_1}{x_2} = -1}$$

→ **Relación entre $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ y $\begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}$**

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}\begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} = \begin{pmatrix} a & \tilde{a} \\ a & -\tilde{a} \end{pmatrix}\begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} \Rightarrow$$

$$\Rightarrow \begin{cases} x_1 = a\eta_1 + \tilde{a}\eta_2 \\ x_2 = a\eta_1 - \tilde{a}\eta_2 \end{cases} \Rightarrow \begin{cases} \eta_1 = \dfrac{x_1 + x_2}{a} \\ \eta_2 = \dfrac{x_1 - x_2}{\tilde{a}} \end{cases} \rightarrow \text{cambio de variable para que las ecuaciones de movimiento se desacoplen.}$$

→ a es arbitrario podemos tomar p. ej. $a = 2 = \tilde{a}$ (normalización)

↳ see Marion page 487